

COTG from Defective Analysis

WDS, Parity Quotients, and Charted Oscillatory Transport Geometry

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Claim boundary. This document defines Charted Oscillatory Transport Geometry (COTG) as a realization layer derived from Defective Analysis through Whole Defect Systems. The WDS path is primary for the derivation. The K_2/K_{odd} parity quotient is a secondary construction route, not the whole definition of COTG. The operational two-sector probe appears only as an example after the main definition. No external application is claimed.

Abstract

COTG is the charted oscillatory realization layer of Defective Analysis. A Whole Defect System supplies an upstream carrier, a defect extractor, defect sectors, a gluing carrier, and a quotient/gluing projection. Under a parity-stable quotient, non-parity defect sectors may be collapsed while a binary gluing kernel is preserved. The analytic COTG layer then studies local oscillatory charts, unitary phase transport, tensor aggregation, gluing residuals, projection, and admissibility. The central separation is between latent phase transport, which preserves magnitude, and projection, which can expose relative phase through interference terms.

1 Position in the DA hierarchy

The hierarchy used in this document is

$$\text{DA} \longrightarrow \text{WDS} \longrightarrow D_P(N) = \Delta N \longrightarrow \text{optional parity quotient} \longrightarrow \text{COTG realization}.$$

COTG is not the whole upstream system. It is a realization layer for charted oscillatory transport, projection, and residual compatibility. The WDS supplies the upstream defect object. COTG supplies the chart-level analytic realization.

2 Whole Defect Systems in DA language

Definition 1 (Whole Defect System). *A Whole Defect System is a tuple*

$$N = (X, \Delta, \{K_p\}_{p \in P}, G, q)$$

where:

- X is the carrier of states before quotient;
- Δ is the defect-extraction operator;
- K_p is a defect sector indexed by an internal label p ;
- G is the gluing or comparison carrier;
- q is the quotient or gluing projection identifying defect directions invisible at the declared resolution.

The WDS is a DA object because it declares a carrier, a defect extractor, a quotient structure, and a comparison carrier. Its validity depends on the category or realization in which these data are interpreted.

Definition 2 (Primal derivative). *The Primal derivative of a WDS is the defect extractor*

$$D_P(N) = \Delta N.$$

It exposes the defect object and the sector data on which quotienting or gluing acts.

Definition 3 (Primal integral). *The Primal integral is reconstruction from a quotient or gluing skeleton, generally only modulo a killed kernel:*

$$I_P(\text{quotient data}) = N \bmod K_{\text{killed}}.$$

It does not reconstruct a unique pre-quotient WDS unless the relevant retained kernel is supplied.

3 The parity quotient as a secondary construction

Some WDS instances admit a sector decomposition

$$\Delta N = \bigoplus_{p \in P} K_p.$$

When one sector is distinguished as a binary or parity sector K_2 , define

$$K_{\text{odd}} = \bigoplus_{p \in P, p > 2} K_p,$$

where $p > 2$ is internal grammar notation for the non-parity sector labels.

Definition 4 (Odd-sector equivalence). *States $x, y \in X$ are odd-sector equivalent when*

$$x \sim_{\text{odd}} y \iff \Delta(x) - \Delta(y) \in K_{\text{odd}}.$$

The quotient is

$$Q_2(N) = X / \sim_{\text{odd}}.$$

Definition 5 (Parity projection). *The induced parity defect projection is*

$$\Delta_2([x]) = \text{proj}_{K_2}(\Delta x).$$

This is well-defined only if odd-sector equivalence does not change the K_2 component:

$$x \sim_{\text{odd}} y \implies \text{proj}_{K_2}(\Delta x) = \text{proj}_{K_2}(\Delta y).$$

This is the parity no-leakage condition.

Proposition 1 (Parity quotient route). *If N is a WDS with a sector decomposition*

$$\Delta N = K_2 \oplus K_{\text{odd}}$$

and if the quotient kills exactly K_{odd} while preserving the K_2 component, then the parity-stable quotient residue is

$$Q_2(N) \simeq N / K_{\text{odd}}$$

in the declared quotient category.

Remark 1. The parity quotient is a construction route. It is not the entire COTG definition. COTG is the charted oscillatory realization of the quotient or of another DA/WDS object satisfying the same admissibility requirements.

4 Finite COTG object

Definition 6 (Finite COTG realization). *A finite COTG realization is a tuple*

$$\mathcal{C} = (M, \mathcal{A}, E, \{X_i\}_{i \in I}, \{q_i\}_{i \in I}, \{g_{ij}\}_{(i,j) \in E}, T, \Pi, R)$$

where:

Symbol	Meaning
M	Index space or base chart domain.
$\mathcal{A} = \{(U_i, \phi_i)\}$	Finite chart family on M .
E	Directed overlap edge set between chart domains.
X_i	Local complex state space or fiber over U_i .
$q_i(t)$	Local oscillatory section on chart U_i .
g_{ij}	Transition-like map on an overlap edge (i, j) .
T	Global tensor or completed state object built from local fibers.
Π	Projection or measurement map into observable space Y .
R	Residual data measuring gluing, compatibility, or projection failure.

The object is admitted by structure, not by label. The chart overlaps, local fibers, transition maps, tensor object, projection, and residual law must all be declared.

5 Local oscillatory charts

For each chart U_i , let X_i be a finite-dimensional complex vector space with basis

$$\{e_{i1}, \dots, e_{iN_i}\}.$$

Definition 7 (Local oscillator). *A local oscillatory section on chart U_i is a finite mode sum*

$$q_i(t) = \sum_{k=1}^{N_i} a_{ik} \exp(i(\omega_{ik}t + \phi_{ik})) e_{ik}.$$

The coefficients a_{ik} determine latent magnitude. The phase terms determine relative transport and projected interference.

6 Phase transport and latent magnitude

Definition 8 (Off-axis phase-time displacement). *An off-axis coordinate α_i determines a displacement*

$$\Delta t_i = \lambda_i \alpha_i.$$

Substitution gives

$$q_i(t + \lambda_i \alpha_i) = \sum_k a_{ik} \exp(i(\omega_{ik}t + \phi_{ik})) \exp(i\omega_{ik}\lambda_i \alpha_i) e_{ik}.$$

Lemma 1 (Local magnitude invariance). *Local phase-time displacement preserves each local mode magnitude.*

Proof. Each coefficient is multiplied by $\exp(i\theta)$ with $\theta \in \mathbb{R}$. Since $|\exp(i\theta)| = 1$, the magnitude is unchanged. $\square$

7 Global tensor state

The finite tensor space is

$$T = \bigotimes_{i \in I} X_i.$$

The shifted global state is

$$Q(t; \alpha) = \bigotimes_{i \in I} q_i(t + \lambda_i \alpha_i).$$

After expansion into a global tensor basis $\{E_k\}_{k \in K}$, write

$$Q(t; \alpha) = \sum_{k \in K} A_k \exp(i(\Omega_k t + \Phi_k + \Theta_k(\alpha))) E_k,$$

where

$$A_k = \prod_i a_{ik_i}, \quad \Omega_k = \sum_i \omega_{ik_i}, \quad \Phi_k = \sum_i \phi_{ik_i}, \quad \Theta_k(\alpha) = \sum_i \omega_{ik_i} \lambda_i \alpha_i.$$

Definition 9 (Latent magnitude object). *The finite latent magnitude object is*

$$M_Q = (|A_k|)_{k \in K}.$$

Theorem 1 (Finite latent magnitude invariance). *For every mode k ,*

$$|A_k \exp(i\psi_k)| = |A_k|.$$

Hence M_Q is independent of t and α under phase-only transport.

8 Gluing residuals

Definition 10 (Transition-like map). *On an overlap edge (i, j) , a transition-like map has the form*

$$\tau_{ij} = \lambda_j \alpha_j - \lambda_i \alpha_i, \quad g_{ij} = F_{ij} \circ S_{\tau_{ij}},$$

where $S_{\tau_{ij}}$ is the phase-time shift component and F_{ij} is the fiber/state component.

Definition 11 (Finite phase residual). *For a matched mode pair (ik, jl) , the phase residual is*

$$R_{ij}^{kl}(\alpha) = \omega_{ik} \lambda_i \alpha_i + \phi_{ik} - \omega_{jl} \lambda_j \alpha_j - \phi_{jl} \pmod{2\pi}.$$

Full gluing also requires amplitude, frequency, and basis compatibility. Thus one may record

$$D_{ij}^{kl} = (A_{ij}^{kl}, W_{ij}^{kl}, B_{ij}^{kl}, R_{ij}^{kl}).$$

9 Projection and observable interference

Let

$$\Pi : T \rightarrow Y$$

be a linear map into a finite-dimensional inner-product space. The projected observable is

$$y(t; \alpha) = \Pi Q(t; \alpha) = \sum_k A_k \exp(i\psi_k(t, \alpha)) v_k, \quad v_k = \Pi(E_k).$$

Theorem 2 (Projected interference formula). *The projected squared norm decomposes as*

$$\|y(t; \alpha)\|^2 = \sum_{k,l} A_k \overline{A_l} \exp(i(\psi_k - \psi_l)) \langle v_k, v_l \rangle.$$

Equivalently,

$$\|y(t; \alpha)\|^2 = \sum_k |A_k|^2 \|v_k\|^2 + \text{off-diagonal interference}.$$

Proof. Expand $\langle y, y \rangle$. Diagonal terms give $|A_k|^2 \|v_k\|^2$. Off-diagonal terms contain relative phases and projected Gram entries $\langle v_k, v_l \rangle$. $\square$

Corollary 1 (Steering criterion). *The projected observable is phase-steerable by α exactly when the off-diagonal interference functional is nonconstant in α .*

10 Arbitrary-dimensional extension

The arbitrary-dimensional version replaces finite vector spaces by topological state spaces, finite sums by convergent spectral series or direct integrals, finite tensor products by completed tensor products or replacement objects, and finite Gram matrices by kernels or quadratic forms.

For each chart index i , one declares a complex topological vector space H_i . A spectral representation may take the form

$$q_i(t) = \sum_{k \in K_i} a_i(k) \exp(i(\omega_i(k)t + \phi_i(k))) e_{i,k}$$

or

$$q_i(t) = \int_{\Sigma_i} a_i(\sigma) \exp(i(\omega_i(\sigma)t + \phi_i(\sigma))) e_{i,\sigma} d\mu_i(\sigma).$$

The phase transport must be a well-defined unitary multiplier on the chosen state class or dense domain:

$$(U_{i,\alpha_i} f)(\sigma) = \exp(i\omega_i(\sigma)\lambda_i\alpha_i) f(\sigma).$$

11 Intrinsic admissibility

A candidate realization is admitted by reconstruction of structure, not by name. It must specify:

Criterion	Requirement
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Base	Chart intersections reproduce the overlap structure or gluing category.
Fiber	Local fibers preserve mode, spectral, or generalized basis data.
Transition	Reconstructed transition diagrams commute on common domains.
Transport	Phase-only transport is represented by unitary multipliers.
Projection	The projection is defined on the transported state family and preserves Gram/kernel data.
Residual	The state class is compatible with gluing, transport, invariants, and projection.
Limit	Finite-to-infinite passage specifies state, magnitude, and observable convergence.
Quotient	If the parity quotient is invoked, odd-sector collapse preserves the parity sector.

12 Operational example: two-sector alpha phase

This example is not the definition of COTG. It illustrates the operational use of the definition.

Let A be a fixed transport operator on a hidden but probeable finite system. The carrier is fixed by the θ -clock. The α -clock shifts the relative phase between two legal probe sectors.

Let

$$z = z_+ + z_-$$

be a decomposition into two legal sectors, and define

$$z(\varphi) = z_+ + e^{i\varphi} z_-, \quad \varphi \in [-\pi, \pi].$$

The scalar response is

$$R_\varphi(k) = z(\varphi)^* A^k z(\varphi).$$

Expanding gives

$$R_\varphi(k) = z_+^* A^k z_+ + z_-^* A^k z_- + e^{i\varphi} z_+^* A^k z_- + e^{-i\varphi} z_-^* A^k z_+.$$

Thus a global phase on all of z would cancel, but a relative phase between sectors can change the projected scalar response. Phase sensitivity is detected when

$$\frac{\partial R_\varphi(k)}{\partial \varphi} \neq 0$$

for some k .

The legal-probe rule is

$$z \in \text{Sector}_{\alpha, \theta}(\mathcal{P}),$$

where $\mathcal{P}$ is the declared measurement-port space. No hidden support is directly accessed. It may only be inferred from scalar response constraints.

13 Failure surfaces

Code	Failure surface
C1	Undefined chart overlaps or gluing category.
C2	Mode mismatch: amplitude, frequency, basis, or spectral compatibility is missing.
C3	Nonunitary phase transport: claimed phase-only motion changes latent magnitude.
C4	Operator-domain failure: projection, transition, or residual operator is undefined on the state.
C5	Divergent expansion: spectral series or integral fails in the declared topology.
C6	Unspecified tensor completion or replacement object.
C7	Projection kernel failure: the Gram/kernel expression is undefined.
C8	False steering: projected interference is constant despite a steering claim.
C9	Uncontrolled finite-to-infinite passage.
C10	Quotient ambiguity: parity quotient is invoked without preserving the parity sector.
C11	COTG/WDS conflation: COTG is treated as the full WDS rather than a realization layer.

14 Known unknowns and proof obligations

1. Fix the ambient category for each WDS instance.
2. Specify the algebraic status of sector decompositions such as $K_2 \oplus K_{\text{odd}}$.
3. Prove no-leakage for any invoked quotient.
4. Declare tensor completions and operator domains in infinite-dimensional examples.
5. Characterize when operational phase-sector probes are minimal and admissible.

15 Final compression

COTG is the DA realization layer for charted oscillatory transport:

local charts + unitary phase transport + gluing residuals + tensor aggregation + projection + admissibility.

The WDS path is

$N \rightarrow D_P(N) = \Delta N \rightarrow \text{sector data} \rightarrow \text{optional parity quotient} \rightarrow \text{COTG realization.}$

The central analytic separation is

latent phase transport preserves magnitude; projection can expose relative phase.